In-class Exercise

Econometrics, Fall 2026

The following table gives the joint probability distribution between employment status and college graduation among those either employed or looking for work (unemployed) in the working-age population of South Africa.

	Unemployed ($Y = 0$)	Employed ($Y = 1$)	Total
Non–college grads ($X = 0$)	0.078	0.673	0.751
College grads ($X = 1$)	0.042	0.207	0.249
Total	0.12	0.88	1.000

a. Compute E(Y)

b. Calculate E(Y|X=1) and E(Y|X=0)

c. Are educational achievement and employment status independent? Explain